CURRICULUM VITAE

BIOGRAPHICAL

Name: Sungjin Ko

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EDUCATION and TRAINING

UNDERGRADUATE

3/2003	Jeonbuk National University (South Korea)	DVM. (2007)	Veterinary Medicine

GRADUATE

3/2007	Seoul National University (South Korea)	Ph.D. (2012)	Veterinary Internal Medicine

POSTGRADUATE

1/2013–4/2016	Department of Developmental Biology, School of Medicine, University of Pittsburgh	Post-doctoral Researcher	Liver regeneration /PI: Donghun Shin
5/2016–9/2017	Department of Developmental Biology, School of Medicine, University of Pittsburgh	Research Associate	Liver regeneration /PI: Donghun Shin
10/2017–12/2017	Department of Pathology, School of Medicine, University of Pittsburgh	Research Associate	Liver regeneration and cancer/PI: Paul Monga

APPOINTMENTS and POSITIONS

ACADEMIC

1/2018–10/2019	Department of Pathology, School of Medicine, University of Pittsburgh	Research Instructor	Liver regeneration and cancer
11/2019–6/2021	Department of Pathology, School of Medicine, University of Pittsburgh	Research Assistant Professor	Liver regeneration and cancer
7/2021–present	Department of Pathology, School of Medicine, University of Pittsburgh	Assistant Professor (Tenure Stream)	Liver regeneration and cancer

9/2025–present	Hillman Cancer Center, School of Medicine, University of Pittsburgh	Assistant Professor	Liver cancer

CERTIFICATION and LICENSURE

MEDICAL or OTHER PROFESSIONAL LICENSURE:

DVM, Veterinary Medical Board (South Korea)	2007

MEMBERSHIP in PROFESSIONAL and SCIENTIFIC SOCIETIES

Honored Korean Scientist of Biological Research Information Center	2015 - Present
Pittsburgy Liver Research Center	2019 – Present
The American Society for Investigative Pathology	2019 – Present

HONORS

National Research Foundation of Korea Grant fund by the Korean Government	2007
Pilot and Feasibility Grant from Pittsburgh Liver Research Center	2019
George K. Michalopoulos Junior Faculty Travel Award from ASIP	2020
Monga-Hans Junior Faculty Scholar Award from ASIP	2022

PUBLICATIONS

1. ORIGINAL PEER REVIEWED ARTICLES (research)

1) Kim M, Park Y, Covitz R, Kwon J, Liu JJ, Liu S, **Sungjin Ko*** SALL4 Is Required for YAP1-dependent Malignant and Regenerative Hepatocyte-to-Cholangiocyte Reprogramming. 2025 Sep. ***Cancer Res Commun.*** **PMID: 40932240**
2) Park Y[#], Hu S[#], Kim M, Oertel M, Singhi A, Monga SP, Liu S, **Sungjin Ko*** Context-Dependent Distinct Role of SOX9 in Combined Hepatocellular Carcinoma-Cholangiocarcinoma. 2024 Aug. ***Cells.*** **PMID: 39273023**
3) Hu S, Molina L, Tao J, Liu S, Hassan M, Singh S, Poddar M, Bell A, Sia D, Oertel M, Raeman R, Nejak-Bowen K, Singhi A, Luo J, Monga SP*, **Sungjin Ko***. NOTCH-YAP1/TEAD-DNMT1 axis drives hepatocyte reprogramming into intrahepatic cholangiocarcinoma. 2022 Aug. ***Gastroenterology***. ***, Corresponding author. PMID: 35550144**
4) Zhang Y, Xu H, Cui G, Liang B, Chen X, **Sungjin Ko**, Affo S, Song X, Liao Y, Feng J, Wang P, Wang H, Xu M, Wang J, Pes GM, Ribback S, Zeng Y, Singhi A, Schwabe RF, Monga SP, Evert M, Tang L, Calvisi DF, Chen X. β-Catenin sustains and is required for Yes-associated

protein oncogenic activity in cholangiocarcinoma. 2022 Aug. ***Gastroenterology***. **PMID: 35489428**

5) Hu S[#], Russell JO[#], Liu S, Cao C, McGaughey J, Rai R, Kosar K, Tao J, Hurley E, Poddar M, Singh S, Bell A, Shin D, Raeman R, Singhi AD, Nejak-Bowen K, **Sungjin Ko*** and Monga SP*. β-Catenin-NFkB-CFTR interactions in cholangiocytes regulate inflammation and fibrosis during ductular reaction. 2021 Oct. ***Elife.*** ***, Corresponding author PMID: 34609282**
6) Molina LM, Zhu J, Li Q, Pradhan-Sundd T, Krutsenko Y, Sayed K, Jenkins N, Vats R, Bhushan B, **Sungjin Ko**, Hu S, Poddar M, Singh S, Tao J, Sundd P, Singhi A, Watkins S, Ma X, Benos PV, Feranchak A, Michalopoulos G, Nejak-Bowen K, Watson A, Bell A, Monga SP. Compensatory hepatic adaptation accompanies permanent absent of intra-hepatic biliary network due to YAP1 loss in liver progenitors. 2021 Jul. ***Cell Report***. **PMID: 34233187**
7) Kyounghwa Jung, Minwook Kim, Juhoon So, Seung-Hoon Lee, **Sungjin Ko*** and Donghun Shin*. Farnesoid X receptor activation impairs liver progenitor cell-mediated liver regeneration via the PTEN-PI3K-AKT-mTOR axis in zebrafish. 2021 Jul. ***Hepatology*****. *, Corresponding author. PMID: 33314176**
8) Pradhan-Sundd T, Lui S, Singh S, Poddar M, **Sungjin Ko**, Bell A, Frank J, Huck I, Stolz D, Apte U, Ranganathan S, Nejak-Boewn K, Monga SP. Dual b-Catenin and g-Catenin loss in Hepatocytes impacts their polarity through altered transforming growth factor-b and hepatocyte nuclear factor 4-a signaling. 2021 May. ***Am J Pathology***. **PMID: 33662348**
9) Chad Walesky, Kellie Kolb, Carolyn Winston, Jake Henderson, Benjamin Kruft, Ira Fleming, **Sungjin Ko**, Satdarshan Monga, Florian Mueller, Udayan Apte, Alex Shalek, and Wolfram Goessling. Functional Compensation Precedes Recovery of Tissue Mass Following Acute Liver Injury. 2020. ***Nature Communications*****.**
10) So J, Kim M, LEE SH, **Sungjin Ko**, Lee DA, Park H, Azuma M, Parsons MJ, Prober D, Shin D. Attenuating the EGFR-ERK-SOX9 axis promotes liver progenitor cell-mediated liver regeneration in zebrafish. 2020. ***Hepatology****.*
11) Adeola Michael*, **Sungjin Ko***, Junyan Tao, Akshata Moghe, Hong Yang, Meng Xu, Jacquelyn O Russell, Tirthadipa Pradhan-Sundd, Silvia Liu, Sucha Singh, Minakshi Poddar, Jayvir S Monga, Pin Liu, Michael Oertel, Sarangarajan Ranganathan, Aatur Singhi, Sandra Rebouissou, Jessica Zucman-Rossi, Silvia Ribback, Diego Calvisi, Natalia Qvartskhava, Boris Görg, Dieter Häussinger, Xin Chen, Satdarshan P Monga. Inhibiting Glutamine-Dependent mTORC1 Activation Ameliorates Liver Cancers Driven by β-Catenin Mutations. 2019. ***Cell Metabolism.*** ***, Co-first authors**
12) Jacquelyn O. Russell[#], **Sungjin Ko[#,*]**, Satdarshan P. Monga, and Donghun Shin*. Notch inhibition promotes differentiation of liver progenitor cells into hepatocytes via sox9b repression in zebrafish. 2019. ***Stem Cells International.*** **[#], Co-first authors; *, Corresponding authors**
13) **Sungjin Ko** and Shin D. Chemical screening using a zebrafish model for liver progenitor cell-driven liver regeneration. 2019. ***Methods Mol Biol.***
14) **Sungjin Ko*,** Russell JO, Tian J, Gao C, Kobayashi M, Feng R, Yuan X, Shao C, Ding H, Poddar M, Singh S, Locker J, Weng HL, Monga SP, Shin D*. Hdac1 Regulates Differentiation of Bipotent Liver Progenitor Cells During Regeneration via Sox9b and Cdk8. 2018. ***Gastroenterology.*** ***, Corresponding author**

15) Khaliq M, **Sungjin Ko**, Liu Y, Wang H, Solnica-Krezel L, Shin D. Stat3 Regulates Liver Progenitor Cell-Driven Liver Regeneration in Zebrafish. 2018. ***Gene Expr.***
16) Russell JO[#], **Sungjin Ko[#]**, Saggi HS, Singh S, Poddar M, Shin D, Monga SP. Bromodomain and Extraterminal (BET) Protein Regulate Hepatocyte Proliferation in Hepatocyte-Driven Liver Regeneration. 2018. ***Am J Pathology.*** **[#], Co-first authors**
17) Jun-gu Kang, Hee-woo Lee, **Sungjin Ko**, and Joon-seok Chae. Comparative proteomic analysis of the outer membrane protein 43 (omp43) deficient Bartonella henselae. 2018. ***J Vet Sci.***
18) Jun-Gu Kang, **Sungjin Ko**, Heung-Chul Kim, Sung-Tae Chong, Terry A Klein, Jeong-Byoung Chae, Yong-Sun Jo, Kyoung-Seong Choi, Do-Hyeon Yu, Bae-Keun Park, Jinho Park, Joon-Seok Chae. Prevalence of Anaplasma and Bartonella spp. in Ticks Collected from Korean Water Deer (Hydropotes inermis argyropus). 2016. ***Korean J Parasitol*.**
19) **Sungjin Ko[#],** Tae-Young Choi[#], Jacqelyn O. Russell, Juhoon So, Satdarshan P.S. Monga, Donghun Shin. Bromodomain and extraterminal (BET) proteins regulate biliary-driven liver regeneration. **Journal of *Hepatology.*** 2016 **[#], Co-first authors**
20) **Sungjin Ko**, JG Kang, HC Kim, TA Klein, KS Choi, JW Song, HY Youn, JS Chae. Prevalence, Isolation and Molecular Characterization of Bartonella Species in Republic of Korea. 2016. ***Transbound Emerg Dis.***
21) Jun-Gu Kang[#], **Sungjin Ko[#],** Barney Smith, Heung-Chul Kim, In-Yong Lee, Joon-Seok Chae. Prevalence of Anaplasma, Bartonella and Borrelia Species in Haemaphysalis longicornis Collected from Goats, Democratic People's Republic of Korea. 2016. ***J Vet Sci*.** **[#], Co-first authors**
22) Choi TY, Khaliq M, **Sungjin Ko**, So J, Shin D. Hepatocyte-specific ablation in zebrafish to study biliary-driven liver regeneration. 2015. ***J Vis Exp*.**
23) HC Kim, **Sungjin Ko**, CY Choi, HY Nam, HY Chae, ST Chong, TA Klein. Migratory bird tick surveillance, including a new record of Haemaphysalis ornithophila Hoogstraal and Kohls, 1959 (Acari: Ixodidae) at Hong Island, Republic of Korea. 2014. ***Systematic and Applied Acarology.***
24) **Sungjin Ko**, Jun-gu Kang, Jung-yong Yeh, Jin-san Moon, Gui-cheol Choi, Sohyun Won and Joon-seok Chae. First Report on Molecular Detection of Equine Upper Respiratory Infectious Viruses in Republic of Korea. 2013. ***J Equine Vet Sci.***
25) **Sungjin Ko**, Su-jin Kim, Jun-gu Kang, Sohyun Won, Hang Lee, Nam-shik Shin, Kyoung–seong Choi, Hwa-young Youn and Joon-seok Chae. Molecular Detection of Bartonella grahamii and B. schoenbuchensis-related Species in Korean Water Deer (Hydropotes inermis argyropus). 2013. ***Vector Borne Zoonotic Dis.***
26) Baek-Jun Kim, Su-Jin Kim, Jun-Gu Kang, **Sungjin Ko**, Sohyun Won, Hyewon Kim, Heung-Chul Kim, Myung-Soon Kim, Sung-Tae Chong, Terry A. Klein, Sanghun Lee, Joon-Seok Chae. First Report for the Seasonal and Annual Prevalence of Flea-Borne Bartonella from Rodents and Soricomorphs in the Republic of Korea. 2013. ***Vector Borne Zoonotic Dis.***
27) Nam EH, **Sungjin Ko**, Chae JS, Hwang CY. Characterization and Zoonotic Potential of Uropathogenic Escherichia coli Isolated from Dogs. 2013. **J Microbiol Biotechnol.**
28) Kang JG, Kim HC, Choi CY, Nam HY, Chae HY, Chong ST, Klein TA, **Sungjin Ko**, Chae JS. Molecular Detection of Anaplasma, Bartonella, and Borellia Species in Ticks Collected from Migratory Birds from Hong-do Island, Republic of Korea. 2013. ***Vector Borne Zoonotic Dis.***

29) Kim HN, Lee JH, Jin WJ, **Sungjin Ko**, Jung K, Ha H, Lee ZH. MS-275, a benzamide histone deacetylase inhibitor, prevents osteoclastogenesis by down-regulating c-Fos expression and suppresses bone loss in mice. 2012. ***Eur J Pharmacol*.**
30) Kim HC, Han SH, Chong ST, Klein TA, Choi CY, Nam HY, Chae HY, Lee H, **Sungjin Ko**, Kang JG, Chae JS. Ticks collected from selected mammalian hosts surveyed in the Republic of Korea during 2008-2009. 2011. ***Korean J Parasitol.***
31) **Sungjin Ko#**, Kim HC#, Yang YC, Chong ST, Richards AL, Sames WJ, Klein TA, Kang JG, Chae JS. Detection of Rickettsia felis and Rickettsia typhi and seasonal prevalence of fleas collected from small mammals at Gyeonggi Province in the Republic of Korea. 2011. ***Vector Borne Zoonotic Dis.* #, Co-first authors**
32) Kang JG, **Sungjin Ko**, Kim YJ, Yang HJ, Lee H, Shin NS, Choi KS, Chae JS. New genetic variants of Anaplasma phagocytophilum and Anaplasma bovis from Korean water deer (Hydropotes inermis argyropus). 2011. ***Vector Borne Zoonotic Dis***.
33) **Sungjin Ko**, Kang JG, Kim SY, Kim HC, Klein TA, Chong ST, Sames WJ, Yun SM, Ju YR, Chae JS. Prevalence of tick-borne encephalitis virus in ticks from southern Korea. 2010. ***J Vet Sci.***
34) Kim HC, Yang YC, Chong ST, **Sungjin Ko**, Lee SE, Klein TA, Chae JS. Detection of Rickettsia typhi and seasonal prevalence of fleas collected from small mammals in the Republic of Korea. 2010. ***J Wildl Dis.***
35) Chae JS, Adjemian JZ, Kim HC, **Sungjin Ko**, Klein TA, Foley J. Predicting the emergence of tick-borne infections based on climatic changes in Korea. 2008. ***Vector Borne Zoonotic Dis.***

2. OTHER PEER-REVIEWED PUBLICATIONS

➢ Commentaries

1) **Sungjin Ko**, Monga SP. Hepatic zonation now on hormones! 2018. ***Hepatology.***

➢ Invited Reviews

2) Cheon I, Kim M, Kim KH, **Sungjin Ko**. Hepatic Nuclear Receptors in Cholestasis-to-Cholangiocarcinoma Pathology. 2025 Mar**.** ***Am J Pathol.*** **PMID: 39326734**
3) Kim M, Park Y, Kim YS, **Sungjin Ko.** Cellular Plasticity in Gut and Liver Regeneration. 2024 Nov. ***Gut Liver.*** **PMID: 39081200**
4) Kim YS, Hurley EH, Park Y, **Sungjin Ko.** Treatment of primary sclerosing cholangitis combined with inflammatory bowel disease. 2023 Oct. ***Intest Res.*** **PMID: 37519211**
5) Kim YS, Hurley EH, Park Y, **Sungjin Ko.** Primary sclerosing cholangitis (PSC) and inflammatory bowel disease (IBD): a condition exemplifying the crosstalk of the gut-liver axis. 2023 July. ***Exp Mol Med.*** **PMID: 37464092**
6) Minwook Kim, Evan Delgado, **Sungjin Ko**. DNA methylation in cell plasticity and malignant transformation in liver diseases. 2023 Jul. ***Pharmacol Ther.*** **PMID: 36535346**
7) **Sungjin Ko***, Kim M, Molina L, Sirica AE, Monga SP*. YAP1 activation and Hippo pathway signaling in the pathogenesis and treatment of intrahepatic cholangiocarcinoma. 2022. ***Adv Cancer Res.* *, Corresponding author. PMID: 35961703**

8) **Sungjin Ko#**, Jacquelyn O. Russell#, Laura Molina, Satdarshan P. Monga. Liver progenitor cells and cell plasticity in liver injury and repair: knowns and unkowns. 2020. ***Annual Review of Pathology.*** **#, Co-first authors**

3. OTHER NON-PEER-REVIEWED PUBLICATIONS

➢ Editorials

1) **Sungjin Ko**, Monga SP. The Dark side of Human Hepatocyte Plasticity. Gastroenterology. 2024 Oct. ***Gastroenterology***. **PMID: 38936779**
2) **Sungjin Ko.** Trailblazing TRAIL Therapy: Illuminating Pathway for Cholangiocarcinoma Treatment. 2024 Feb. ***Cell Mol Gastroenterol Hepatol.*** **PMID: 38431266**

4. BOOKS, BOOK CHAPTERS AND MONOGRAPHS

5. PUBLISHED ABSTRACTS (in Scientific Journal)

6. ABSTRACTS (not published in Scientific Journals)

1) **Sungjin Ko**, Monga J, Krutsenko Y, Liu S, Hu S, Singh S, Poddar M, Bell AW, Luo JH, Moghe A, Singhi A, Jurczak M, Monga SS. Long-term consequences of liver regeneration. 2022. ***Hepatology.***
2) Hu S, Molina L, Tao J, Liu S, Hassan M, Singh S, Poddar M, Nejak-Bowen K, Oertel M, Raeman R, Singhi A, Bell A, **Sungjin Ko**, Monga SP. Therapeutic implication of SOX9 and YAP1 co-repression in advanced intrahepatic cholangiocarcinoma. 2022. ***FASEB J.***
3) Molina LM, Gabdulkhakova A, Zhu J, Liu S, Poddar M, Singh S, Ma X, Singhi A, **Sungjin Ko**, Monga SP. In the absence of YAP, TAZ contributes to hepatocyte adaptation in chronic cholestasis in females. 2022. ***FASEB J.***
4) Monga SP, **Sungjin Ko**, Hu S, Singh S, Poddar M. Single-cell spatial transcriptomics analysis of a regenerating liver. 2022. ***FASEB J.***
5) Hu S, Russell JO, Liu S, Rai R, Hurley E, Poddar M, Singh S, Bell A, Raeman R, **Sungjin Ko**. Lack of β-catenin in cholangiocytes regulates immune environment and wound healing after hepatobiliary injury. 2020. ***Am J Pathol.***
6) **Sungjin Ko**, Hu S, Molina L, Tao J, Liu S, Hassan M, Singh S, Poddar M, Nejak-Bowen K, Oertel M. Simultaneous YAP and SOX9 repression eliminates Notch-dependent hepatocyte-driven cholangiocarcinogenesis. 2020. ***Am J Pathol.***
7) **Sungjin Ko**, Hu S, Tremblay JM, Erasmus J, Talavera JC, Shoemaker CB, Kansra V, Monga SP. Highly efficient single-domain antibody neutralization of interleukin-6, the factor at the epicenter of cytokine storm in acutely ill COVID-19 patients. 2020. ***Am J Pathol.***
8) Molina LM, Zhu JJ, Li Q, Pradhan-Sundd T, Sayed K, Jenkins N, **Sungjin Ko**, Hu S, Poddar M, Ma X, Monga SP. Hepatoblast loss of YAP1 leads to Alagille syndrome-like phenotype and severe chronic kidney disease. 2020. ***Am J Pathol.***
9) Hu S, Russell JO, Liu S, Rai R, Kosar K, Hurley E, Poddar M, Singh S, Bell AW, Raeman R, **Sungjin Ko**. Lack of β-catenin in cholangiocytes regulates immune environment and wound healing after hepatobiliary injury. 2020. ***Am J Pathol.***
10) **Sungjin Ko**, Hu S, Tremblay JM, Erasmus J, Talavera JC, Shoemaker CB, Kansra V, Monga SP.

Highly efficient single-domain antibody neutralization of interleukin-6, the factor at the epicenter of cytokine storm in acutely ill COVID-19 patients. 2020. ***Am J Pathol*.**

11) Hu S, **Sungjin Ko**, Kim MW, Preziosi M, Bell A, Monga S. Elimination of Wnt secretion from hepatic stellate cells impairs hepatocyte proliferation after partial hepatectomy. 2020. ***FASEB J*.**
12) **Sungjin Ko**, Molina L, Tao J, Liu S, Hassan M, Hu S, Nejak-Bowen K, Oertel M, Raeman R, Singhi A. Simultaneous Yap and Sox9 repression eliminates Notch-dependent hepatocyte-driven cholangiocarcinogenesis. 2020. ***FASEB J*.**
13) Walesky C, Kolb K, Winston C, Henderson J, Kruft B, Fleming I, **Sungjin Ko**, Monga SP, Mueller F, Apte U. Functional compensation precedes recovery of tissue mass following acute liver injury. 2020. ***FASEB J*.**
14) Monga S, **Sungjin Ko**, Molina L, Tao J, Singhi A, Bell A. Hepatocyte-derived intrahepatic cholangiocarcinoma requires Yap and Sox9: A clinical and preclinical analysis. 2020. ***J Clin Oncol*.**
15) **Sungjin Ko**, Monga SP. Elimination of Wnt secretion from hepatic stellate cells impairs cyclin-D1 at baseline and hepatocyte proliferation after partial hepatectomy. 2020. ***Hepatology***.
16) **Sungjin Ko,** Monga SP. Elucidating the therapeutic effect of SOX-9 and/or YAP elimination in intrahepatic cholangiocarcinoma. 2020. ***Hepatology***.
17) **Russell JO, Sungjin Ko**, Shin DH, Monga SP Bromodomain and Extra-terminal (BET) proteins regulate hepatocyte proliferation in hepatocyte-driven liver regeneration. 2020. *FASEB Journal*.
18) **Sungjin Ko, Russell JO, Shin D, Monga SP.** Roles for Notch-Sox9 in biliary epithelial cells-to-hepatocytes differentiation during liver regeneration. 2019. ***FASEB J*.**
19) **Sungjin Ko,** Kim MW, Preziosi ME, Monga SP. Elimination of Wnt secretion from hepatic stellate cells impairs hepatocyte proliferation after partial hepatectomy. 2020. *FASEB J*.
20) **Sungjin Ko,** Jung K, Russell JO, Kim MW, So J, Monga SP, Shin D. FXR activation impairs liver progenitor cell-driven liver regeneration by enhancing PTEN-dependent inhibition of AKT-mTOR axis. 2019. ***FASEB J*.**
21) Tae-Ho Kim, Dong-Ha Bhang, **Sungjin Ko**, Hee-Jeong, Youn, Joon-Seok Chae, Hee-Woo, Lee, Hwa-Young Youn, Suh-Young Sohn, Ye-rin, Coh. Recombinant Bartonella adhesion A stimulates inflammatory response and angiogenesis in human umbilical vein endothelial cells. 2019. ***J Vet Internal Med.***
22) Russell JO, **Sungjin Ko**, Shin D, Monga SP. Bromodomain and Extra-terminal (BET) proteins regulate hepatocyte proliferation in hepatocyte-driven liver regeneration. 2018. ***Am J Pathol*.**

7. PUBLICATION EQUIVILENT SCHOLARLY PRODUCTS

PROFESSIONAL ACTIVITIES

TEACHING

Lecture and Course

Year	Graduate Course Name	Role	# of graduate students	Total # of lectures in course (my lecture)	Contact hours
2023	Cancer Biology and Therapeutics	co-director with Drs. Evan Delgado and Laura Stabile	13 graduate students	32 (1)	5
2024	Cancer Biology and Therapeutics	co-director with Drs. Evan Delgado and Laura Stabile	31 graduate students	32 (1)	8

Graduate Student Teaching/Mentoring:

	Year	Type of learner	Name	Achievement
1	2018-2021	MD/PhD student	Laura Molina	2 co-authored publications (original research)
2	2018	Visiting PhD student	Florencia Lorenzetti	
3	2018	Rotation student (ISB program)	Minwook Kim	
4	2019-2021	Visiting PhD student (Tsinghua University)	Shikai Hu	4 co-authored publications i.e., 3 first-corresponding author publications (original research)
5	2021	Rotation student	Joud Mulla	
6	2021	PhD student	Yekaterina Krutsenko	
7	2022	Undergraduate student	Anthony Zhu	
8	2022	Visiting MS	Yoojeong Park	5 co-authored publications i.e., 2 original research and 3 invited reviews
9	2023	MD/PhD student	Brandon Lehrich	
10	2023	MD/PhD student	Tyler Yasaka	
11	2025	Rotation PhD student (IBGP program)	Rachel Covitz	1 co-authored publication

Postdoc Mentoring:

	Year	Type of learner	Name	Achievement
1	2021-2023	Visiting scholar (MD/PhD)	Yoo Sun Kim	3 co-authored publications i.e., 3 first-corresponding author publications (invited review)
2	2023-	Postdoc (PhD)	Minwook Kim	5 co-authored publications i.e., 4 first-corresponding author publications (2 original research and 3 invited review)

Graduate committee

<u>PhD qualifying exam committees:</u>

1. Angela Paul: Molecular Genetics and Developmental Biology (MGDB) Program, 2023 (advisor: Michael Tsang)
2. Brandon Lehrich: MSTP program, 2023 (advisor: Paul Monga)

3. Tyler Yasaka: MSTP program, 2024 (advisor: Paul Monga).
4. Chun Cheng (Tony) Chiang: Interdisciplinary Biomedical Graduate Program (IBGP) program, 2024 (advisor: Kari Nejak-Bowen)

PhD thesis committees:
1. Brandon Lehrich: MSTP program, 2023
2. Tyler Yasaka: MSTP program, 2024 (advisor: Paul Monga)
3. Chun Cheng (Tony) Chiang: Interdisciplinary Biomedical Graduate Program (IBGP) program, 2024 (advisor: Kari Nejak-Bowen) **Chair of the committee*

RESEARCH

Current Grant Support:

1. 1R01 CA300059
(Ko, PI) 04/01/2025-03/31/2030 3.6 calendar months
NIH, NCI Yearly funding: Total-$518,219; Direct-$329,168; Indirect-$189,051
"Deciphering the Molecular Mechanisms Underlying Cholangiocarcinoma Development"

2. 1R01 CA258449
(Ko, PI) 04/01/2021-03/31/2026 3.6 calendar months
NIH, NCI Yearly funding: Total-$323,497; Direct-$203,457; Indirect-$120,040
"Identifying therapeutic options for intrahepatic cholangiocarcinoma"

3. (Ko, MPI) 06/01/2024-12/31/2025 0.6 calendar months
Innovation in Cancer Informatics Discovery Grant
Yearly funding: Total-$25,000; Ko: Direct -$6,000; Indirect-$0
"Discovery and Valida1on of Cancer Therapeutic Targets Through the Single-Cell Long-Read RNA-Seq Analysis"

4. (Ko, Co-I) 07/01/2024-06/31/2026 0.6 calendar months
Competitive Medical Research Fund
Yearly funding: Total-$30,000; Sub-award: Direct-$15,095; Indirect-$0
"Identifying isoform-specific cancer therapeutic targets through the long-read RNA-seq analysis"

5. (Ko, MPI) 09/01/2025-8/31/2027 0.6 calendar months
Incubator Challenge
Yearly funding: Total-$50,000; Ko: Direct -$20,000; Indirect-$0
"Hepatic LCH pathology"

Pending Grant Support:

1. (Ko, Co-I 0.36 calendar months)
* Title: Hepatocyte reprogramming as an optimal regenerative strategy for chronic biliary injury
* Status of Support: Pending
* Name of PD/PI: Kari Nejak-Bowen
* Source of Support: NIH
* Project/Proposal Start and End Date: 7/2026-6/2030
* Total Award Amount (including Indirect Costs): $3,163,961

2. (Ko, Co-I 0.6 calendar months)
* Title: Modeling transcript diversity with bulk and single-cell long-read RNA-seq

* Status of Support: Pending
* Name of PD/PI: MPI: Shuchang Liu/ Pedro Baldoni
* Source of Support: NIH
* Primary Place of Performance: University of Pittsburgh
* Project/Proposal Start and End Date: (MM/YYYY) (if available): 04/2026 – 03/2030
* Total Award Amount (including Indirect Costs): $1,559,660

3. (Ko, Co-I 0.6 calendar months)
*Title: CAR as a Determinant Factor for Intrahepatic Cholestasis-Driven Cholangiocarcinoma
*Status of Support: Pending
* Project Number: Pending
* Name of PD/PI: Kim, Kangho
* Source of Support: DOD/USAMRAA
* Project/Proposal Start and End Date: (MM/YYYY) (if available): 5/1/2026-4/30/2029
* Total Subaward Amount (including Indirect Costs): $39,186

4. (Ko, Co-I 0.6 calendar months)
Title: CFIm loss is a new way to promote cancer resistance by YAP/TAZ activation and disruption of microRNA regulation
*Status of Support: Pending
* Project Number: 1512511
* Name of PD/PI: Hun-Way Hwang
* Source of Support: ACS
* Project/Proposal Start and End Date: (MM/YYYY) (if available): 4/1/2026-3/31/2030
* Total Award Amount (including Indirect Costs): $946,000

5. (Ko, PI 0.6 calendar months)
*Title: Humanized Cholangiocarcinoma Models for Precision Targeting of the SALL4–BMI1 Pathway and Enhancing Immunotherapy in YAP1-Driven CCA
*Status of Support: Pending
Name of PD/PI: Ko, Sungjin
*Source of Support: DOD/USAMRAA
Project/Proposal Start and End Date: (MM/YYYY) (if available): 9/1/2026 – 8/31/2028
* Total Award Amount (including Indirect Costs): $140,000

6. (Ko, PI 0.6 calendar months)
*Title: Modeling BRAFV600E-Driven Hepatic Langerhans Cell Histiocytosis Using Human iPSC-Derived In Vitro and In Vivo Platforms
*Status of Support: Pending
* Name of PD/PI: Ko, Sungjin
*Source of Support: DOD/USAMRAA
* Project/Proposal Start and End Date: (MM/YYYY) (if available): 9/1/2026 – 8/31/2028
* Total Award Amount (including Indirect Costs): $140,000

Prior Grant Support:

1. PF 2019-05 (Ko, PI) 9/1/2018-8/31/2020 1.2 calendar months
Pilot and Feasibility Award from Pittsburg Liver Research Center $ 25,000.00 X 1 year
"Elucidating the therapeutic effect of Sox9 and/or YAP inhibition in intrahepatic cholangiocarcinoma**"**

2. KRF-2007-511-E00042 (Ko, PI) 09/01/2007-08/30/2008 3.6 calendar months

National Research Foundation of Korea $23,000 X 1 year
"Detection of equine upper respiratory infectious viruses by PCR and serological tests"

NON-FUNDED RESEARCH:

➢ Studies:

1. AI/ML-Assisted Histologic Analysis of Clinical Cholangiocarcinoma
Role: MPI (with Dr. Chris Chiu)
Description: Develop automated deep-learning pipelines for diagnostic and prognostic classification of CCA histopathology

2. Bile Duct Humanization for a Preclinical Cholestasis Model
Role: Principal Investigator
Description: Establish a bile-duct–humanized mouse model to study cholangiopathies and drug-induced cholestasis

3. Development of a Humanized Liver Cancer Model
Role: Principal Investigator
Description: Generate human-cell–originated liver cancer models with a humanized immune system in mice for translational and drug-response studies

4. Spatial Transcriptomics of Mixed/Combined HCC-CCA (with Biostatistics)
Role: Multiple Principal Investigator
Status: 6 UPMC cases collected; selecting 2 for Xenium. KRASG12D; p53−/− cHCC-CCA mouse model established; ROIs being selected for Xenium.
Plan: Cross-species comparison of spatial omics (human vs mouse)

Other research related activities:

➢ Editorial Borad:

1. *BMC Medicine* 2025-

➢ AD HOCK AND RECURRING REVIEWS FOR PEER REVIEW JOURNALS (Alphabetical order):

1. *Advanced Science*
2. *American Journal of Pathology*
3. *BMC Cancer*
4. *BMC Medicine*
5. *Cancer Research Communications*
6. *Cell & Bioscience*
7. *Cell Biology and Toxicology*
8. *Cell Communication and Signaling*
9. *Cell Cycle*
10. *Cell Reports*
11. *Cell Transplantation*
12. *Cellular Oncology*
13. *Cells*
14. *Clinical and Experimental Gastroenterology*
15. *Clinical and Molecular Hepatology*
16. *Clinical and Translational Medicine*

17. *CMGH*
18. *Experimental Biology and Medicine*
19. *Experimental and Molecular Medicine*
20. *Gut*
21. *Hematoma Research*
22. *Hepatic Medicine*
23. *Hepatology*
24. *Hepatology Communications*
25. *Human Molecular Genetics*
26. *International Journal of Biological Sciences*
27. *International Journal of Cancer*
28. *International Journal of Molecular Sciences*
29. *Journal of Clinical Investigations*
30. *Journal of Hepatology*
31. *Journal of Hepatology Reports*
32. *Laboratory Investigation*
33. *Liver International*
34. *Molecular Oncology*
35. *Nature Communications*
36. *Oncotargets and Therapy*
37. *Open Biology*
38. *PNAS*
39. *PNAS Nexus*
40. *Pharmacology and Therapeutics*
41. *Science Advances*
42. *Scientific Data*
43. *Scientific Report*
44. *Science Translational Medicine*
45. *Seminars in Liver Diseases*
46. *Stem Cells International*
47. *Theragnostic*
48. *World Journal of Surgical Oncology*

➢ Grant review:

1. Deutsche Forschungsgemeinschaft (DFG) @ 2024
2. DOD Rare Cancers Research Program (RCRP) @ 2024
3. University of Pittsburgh Clinical and Translational Science Institute @ 2025

LIST OF CURRENT RESEARCH INTERESTS

- Liver progenitor cell-driven liver regeneration
- Intrahepatic Cholangiocarcinoma
- Epigenetic changes in cell fate decision
- Chemical screening

- Biomarker-based precision medicine
- Langerhans Cell Histiocytosis–mediated cholangiopathy

INVITED SEMINARS AND LECTURESHIPS

Local Presentations

Date	title of session	sponsoring institution	location
7.27.2025	PMI seminar @ UT Health	UT Health–PMI	Houston
3.20.2024	PSI seminar @ USC	USC–PSI	LA
7.13.2022	SURP seminar	CMP–SURP	Pittsburgh
4.9.2021	G.I Research round	Division of Gastroenterology, Hepatology and Nutrition, University of Pittsburgh	Virtual
9.28.2021	Regenerative Medicine SIG: Targeting hepato-biliary differentiation and function in liver health and disease	Pittsburgh Liver Research Center	Pittsburgh

National Presentations

Date	title of session	sponsoring institution	location
11.9.2020	Host Protection and Responses to Infectious Diseases	PISA 2020	Virtual

International Presentations

Date	title of session	sponsoring institution	location
8.16.2023	Molecular Drivers, Microenvironment and Precision Medicine	FASEB SRC: The Cholangiocarcinoma Conference	Palm Springs, California
4.3.2022	Liver pathobiology: Therapeutics and pathway targets	EB 2022	Philadelphia
4.23.2021	Hepatobiliary Cancer Biomarkers and Molecular Subtypes	Keystone eSymposia (Hepatobiliary Cancers: Pathobiology and Translational Advances)	Virtual
6.29.2021	Mechanisms of Hepatic Tumorigenesis	EB 2021	Virtual

SERVICE

1. Co-director with Dr. Amanda Clark, Summer Undergraduate Researcher Program (SURP) 2023-present